Detectability and Warning Value of Prompt Elasto-Gravity Signals from Manila Trench Megathrust Earthquakes

[**PENDING: author list, station operators, and warning partners**]

Methods-first draft; not for submission

Abstract

We evaluate when prompt elasto-gravity signals (PEGS) from Manila Trench megathrust scenarios can be detected and used for reliable earthquake characterization under realistic South China and surrounding station geometry, non-Gaussian continuous noise, high-microseism and typhoon conditions, station outages, and explicit operational false-alarm constraints. The study compares single-station and coherent physical baselines with a conditional graph model, and quantifies decision-time gain relative to P-wave, W-phase, and GNSS characterization using scenario- and location-specific tsunami arrivals. [**PENDING: Insert registered simulator, station, noise, baseline, and holdout results.**]

1 Introduction

PEGS theory, empirical-noise graph models, and a simulated real-time tsunami- warning implementation already exist. The question addressed here is narrower: whether South China and surrounding networks can provide reliable Manila Trench decisions at a fixed operational false-alarm rate, and what station/noise requirements define the feasible boundary.

2 Scenario and station data

Each source record binds segment, Mw, strike/dip/rake, depth, rupture dimensions, slip, rise time, rupture velocity, and publication. Every tsunami arrival is nested under that source and has a location and arrival definition. Qualitative statements such as "more than about three hours" are not converted into exact universal arrival times.

The open Zhao and Niu scenario descriptor [8] separates southern (13.0–14.5°N) and northern (14.5–22.0°N) Manila source families, with 1000-year recurrence-family magnitudes M_w 8.1 and 9.1. Its 1,800 0.1-degree Slab2.0 subfaults provide location, depth, strike and dip, and the hazard calculation assumes 90-degree rake. The released data are target-point height/period statistics, not per-scenario PEGS sources: rise time, rupture velocity, complete rupture geometry and location-specific arrivals remain missing. These source families are not promoted to simulator inputs.

Liu et al. [6] provide a complementary PMEL/SIFT-based set of 19 finite-fault families: eleven M_w 7.5, five M_w 8.1 and three M_w 8.5 locations, with reported rupture dimensions, mean slip, rake and individual static geometry. Their arrival ranges remain tied to magnitude family and site (Nanao, Shanwei, Macao, Boao or Sanya). Rise time and rupture velocity are not reported, so these records also remain family evidence rather than PEGS-ready dynamic scenarios.

Station candidates require a contemporaneous BH/LH three-component epoch, full response, waveform availability, licence, latency class, and noise quality. Channel metadata or scalar sensitivity alone is insufficient. Noise windows are frozen by season, day/night, calm sea, typhoon/microseism, urban/remote setting, and outage state before model fitting.

The archive/response inventory is retrieved through EarthScope FDSN services [3]. Network-specific citations and acknowledgements remain mandatory; archive availability is never relabelled as real-time latency.

3 Methods

3.1 PEGS simulation benchmark

The selected simulator must reproduce a published event at specified stations and components, with predeclared waveform correlation, amplitude, and onset tolerances, before Manila scenarios are generated. Earth model, reference frame, source-time function, units, sampling, and code commit are recorded.

3.2 Splits and domain shift

Training, validation, and test sets separate rupture segments and noise dates. Domain-out sources and stations are retained. Simultaneous station noise windows preserve spatial correlation. Synthetic white noise is an ablation, not the primary evaluation.

3.3 Physical baselines

Baselines proceed from single-station energy to explicit phase/time-aligned multi-station coherent stacking, template/likelihood detection, and rapid source parameter inversion. Station weights are explicit and L2-normalized. Fixed 0%, 20%, and 40% outage masks retain exact station identities.

3.4 False alarms and detection probability

Noise-only scores calibrate a threshold in false alarms per 30-day month. If the record is too short to resolve one exceedance at the target rate, the threshold is placed above the observed maximum and the resolution limit is reported. Held-out detection probability is then evaluated versus time. Earliest reliable detection may require multiple consecutive qualifying time points.

3.5 Reliable magnitude and warning value

At every time we report magnitude MAE and bias, sensitivity for the predeclared high-risk class (nominally $M_w \geq 8.2$), lower-risk specificity, and prediction- interval coverage. Reliable magnitude time is the first sustained point meeting all criteria.

For location x,

$$\Delta t_{\text{characterization}} = t_{\text{conventional,reliable Mw}} - t_{\text{PEGS,reliable Mw}}, \tag{1}$$

$$T_{\text{lead}}(x) = t_{\text{tsunami arrival}}(x) - t_{\text{PEGS,reliable Mw}}. \tag{2}$$

Negative characterization gain is retained as a negative result.

3.6 Network optimization and conditional GNN

Existing, idealized, and incrementally augmented networks are compared by detection time/probability, false alarms, magnitude error, robustness, and station count. A graph model is trained only if held-out real-noise physical baselines demonstrate stable information. It must outperform those baselines under identical splits and thresholds, not merely fit synthetic waveforms.

4 Results

A registered open-archive audit exact-matches five three-component LH station epochs to complete response structure: HK.HKPS, IC.QIZ, IU.DAV, MY.KKM and TW.KMNB [4, 2, 1, 7, 5]. Their common audited archive interval begins in 2020. The frozen outage design contains one five-station, five four-station and ten three-station combinations for nominal 0%, 20% and 40% branches.

An initial real-noise pilot requested three simultaneous six-hour windows. Thirteen of fifteen waveform requests and twelve response-removal records pass; the failures are retained as two KKM 404 responses and one KMNB gap/overlap case. A failure-triggered replacement adds one complete five-station window. Across the two complete unclassified windows, maximum absolute off-diagonal vertical correlation is 0.039 and 0.311, respectively. These are descriptive archive/response-QC results. No window is yet authorized as quiet calibration data, and no PEGS signal was evaluated. **[PENDING: Insert detection, magnitude and warning results only after the simulator benchmark and environment-labelled noise gates pass.]**

A predeclared USGS event-arrival and IBTrACS regional-track screen passes three of five pilot windows. The Yagi stress window contains two regional earthquake candidates and a Yagi overlap; the February 2025 replacement contains two Philippine earthquake candidates. All five remain unclassified pending local weather, sea-state, microseism and waveform-transient review, so none enters threshold calibration or held-out false-alarm evaluation.

A separate seasonally distributed eight-day acquisition retrieves 38 of 40 station-days and passes response/gap QC for 35. Every day retains at least four stations, but only three have the complete five-station network. The same predeclared catalogue screen retains earthquake candidates in seven days; the remaining day still lacks weather, sea-state and transient labels. A one-shot SeedLink query returns current vertical packets for three of five stations, with packet ages from 94 to 720 s, while two streams return query errors. These archive-yield and endpoint diagnostics neither resolve a 30-day false-alarm rate nor establish operational latency.

A registered seasonal extension requested 40 station-days on eight calendar-selected dates. Thirty-eight were retrieved and 35 passed channel, gap/overlap and response-removal QC. All five failures belong to KKM, leaving only three complete five-station days. Maximum absolute off-diagonal vertical correlation on those unclassified days ranges from 0.016 to 0.308. The frozen response-QC promotion gate therefore fails, and the dates are not replaced from their observed amplitudes. Earthquake screening flags seven of eight dates; none is yet authorized as quiet after all environmental layers. A separate one-shot SeedLink query received current LHZ packets from HKPS, DAV and KKM, while QIZ and KMNB returned query errors. This packet-recency snapshot does not establish uptime or operational latency.

5 Discussion

A high-value negative result is an explicit station-count, placement, noise, and magnitude boundary. Preceding tsunami arrival alone does not establish decision value; the system must characterize hazard reliably and rarely enough to be operationally usable.

6 Data and code availability

Station metadata, waveform licences, exact noise dates, simulator/code versions, scenario provenance, splits, thresholds, compact metrics and checksums will be recorded. Restricted waveforms and large synthetic corpora remain outside Git.

7 Limitations

Archive availability is not real-time latency. The present acceleration spectra use a diagnostic 0.005–0.05 Hz band for response QC, not a frozen PEGS detection passband. Candidate calibration/held-out labels still require earthquake, weather, microseism and transient audits. The simulator benchmark, physical detectability boundary, network performance, warning value and every graph-model stage remain pending. The extracted source families also lack per-scenario rise time, rupture velocity, complete geometry and location-specific arrivals.

References

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